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## Shock Metamorphism



Tianqi Xie and Gordon R. Osinski
Department of Earth Sciences/Institute for Earth and Space Exploration, University of Western Ontario, London, ON, Canada

### Definition

Shock metamorphism is the production of irreversible chemical or physical changes in target materials as a result of the passage of shock waves generated from hypervelocity impact events.

### Introduction

Natural shock metamorphism is the physical consequence of hypervelocity impact of a projectile (e.g., asteroids and comets) into solid target rock, sediment, and regolith exposed on planetary bodies (Stöffler et al. 2018). During passage of the shockwave generated from the impact event, the increase in internal energy accompanying shock compression and subsequent rarefaction results in a unique set of irreversible deformation effects, referred to as shock metamorphic effects, as well as melting and/or vaporization of a volume of target material close to the point of impact (French and Koeberl 2010; Ferrière and Osinski 2013).

Detailed investigation of shock metamorphic effects in geologic materials dates back to 1950s (e.g., Chao et al. 1960, Chao 1967); although the glassy plagioclase phase, maskelynite or diaplectic glass, was first described in the Shergotty achondritic meteorite by Tschermak (1872). Later, shock metamorphic effects, or also known as shock features, have been studied in many minerals and rocks from terrestrial impact structures, lunar samples returned by Apollo and Luna missions, meteorites from the Moon, Mars, Vesta, and many other types of asteroids. Common shock features include shatter cones, undulatory extinction, mosaicism, planar deformation features (PDFs), partial to complete isotropization (diaplectic glass), and high-pressure polymorphs (e.g., von Engelhardt et al. 1970; French and Koeberl 2010; Ferrière and Osinski 2013; Fritz et al. 2017; Stöffler et al. 2018).

In addition to natural shock metamorphism, shock features have also been observed and reproduced in nuclear explosions and in the laboratory through static and dynamic shock recovery experiments. These irreversible changes are produced when rocks and minerals are subjected to shock pressures above their Hugoniot elastic limit (HEL). This limit is defined as "the critical shock pressure at which a solid yields under the uniaxial strain of a plane shock wave" (Stöffler 1972), below which dynamic compression does not produce permanent structural changes; further discontinuities in the Hugoniot curve above the HEL lead to high-density polymorphs. The

© Springer Nature Switzerland AG 2022
B. Cudnik (ed.), *Encyclopedia of Lunar Science*,
https://doi.org/10.1007/978-3-319-05546-6_189-1

shock process results in an increase in internal energy that is partly retained as residual heat on decompression (Stöffler 1972, 1974). The pressure-temperature ranges these features occurred are much higher than those of explosive volcanism or tectonic deformation, make shock metamorphism features the diagnostic of impact events as no other geologic process is capable of producing the extremely high transient pressures and temperatures (Grieve et al. 2014).

It is beyond the scope of this article to discuss the fundamental physics of shock wave propagation in minerals and rocks in detail. With an emphasis on plagioclase, this article aims to provide an overview of shock metamorphic features observed in common rock-forming minerals on the surface of the Moon, modern analytical methods, and progressive stages of shock metamorphism for plagioclase-rich mafic rocks.

## Shock Features

The presence of shock features is the essential indicator for the recognition and confirmation of impact structures on planetary bodies. Minerals subjected to shock metamorphism occur in different petrographic assemblages and in different rock types. The extent and exact nature of the shock features depends on the lithology, mineralogical and microstructural nature of the target rock, and is a function of the magnitude of the hypervelocity impact and of the level of erosion of the crater (French and Koeberl 2010; Ferrière and Osinski 2013).

A great diversity of shock features has been described in the literature during the last 50 years, mostly for quartz, feldspars, olivine and pyroxene (e.g., French and Short 1968; Rubin et al. 1997; Langenhorst 2002; Fritz et al. 2017; Stöffler et al. 2018). The formation of shock features in minerals are largely depends on the crystal structure and on the composition, where tectosilicates are the most sensitive and display the most petrographic deformation effects (Stöffler 1972; French 1998; Langenhorst 2002). Less is known about the effects of shock in other minerals, such as the nesosilicates, double-chain inosilicates, phyllosilicates and carbonate and sulfate minerals in general (Ferrière and Osinski 2013).

Importantly, for lunar crustal materials, all types of lunar samples returned to Earth by the Apollo and Luna missions have been affected by impact processes to some degree. The rocks collected from lunar highlands show an abundance of the anorthosite (>90% anorthite, Ca-rich plagioclase), and the mare basalts are mineralogically comprised of pyroxene, plagioclase, oxides (primarily ilmenite), and olivine (Papike et al. 1998). With the aid of optical microscopy, shock features observed in these common minerals include undulatory extinction, mosaicism, planar fractures, (PFs), planar deformation features (PDFs), and partial to complete isotropization (diaplectic glass). These shock features are described below.

### Shatter Cones

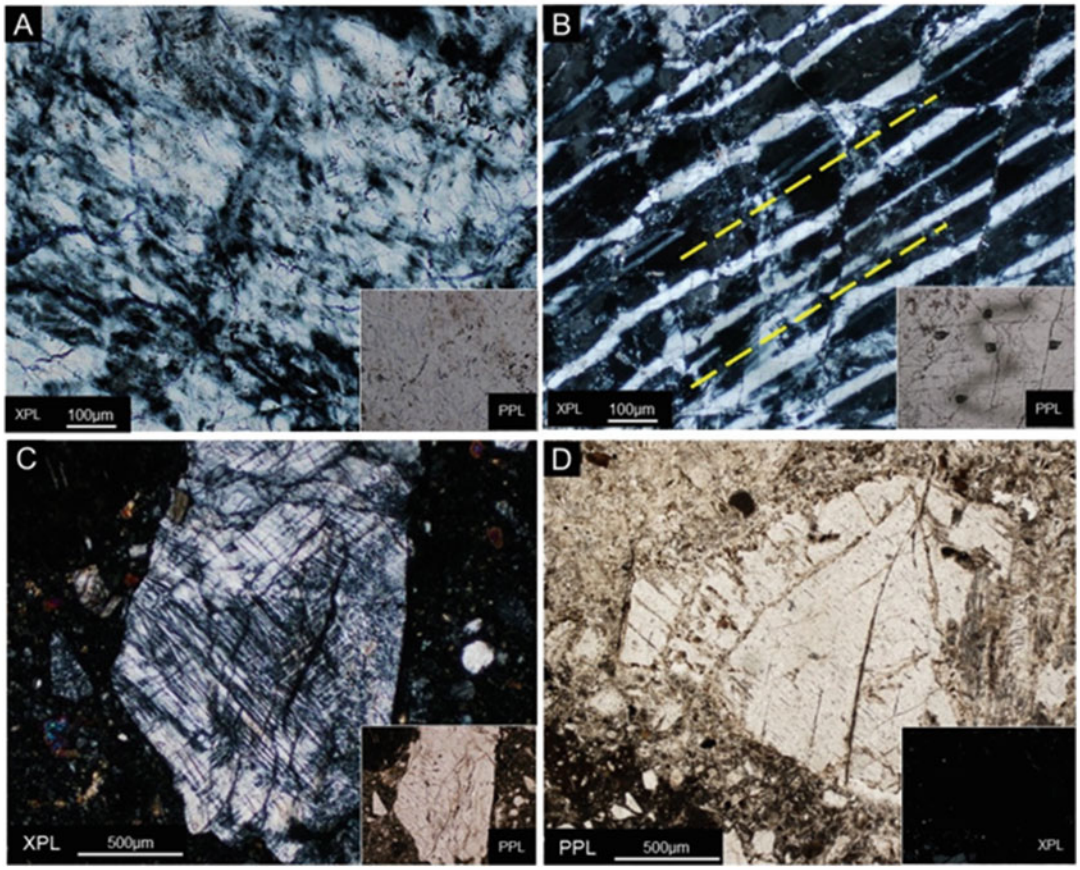
Shatter cones are 'distinctive curved, striated fractures that typically form partial to complete cones' (Dietz 1959; French 1998). Shatter cones are the only diagnostic evidence of hypervelocity impact that can be seen with the naked eye (French and Koeberl 2010). These meso- to macro-scale features, consisting of conical striated fracture surfaces, are best developed in fine-grained lithologies but can also be observed – albeit more poorly developed – in coarser-grained lithologies. They should, in theory, be present on the Moon as shatter cones are found in basalt and anorthosite in terrestrial craters (Baratoux and Reimold 2016; Osinski and Ferrière 2016).

### Undulatory Extinction

Undulatory extinction refers to the appearance of a slightly strained birefringent crystal in cross-polarized transmitted light. A crystal showing undulose extinction has somewhat non-uniform extinction caused by slight-to-moderate misorientation of the crystal lattice as a result of plastic deformation (Stöffler et al. 1988), as in Fig. 1a.

### Mosaicism

Mosaicism refers to the appearance of an intensely strained birefringent grain with an irregular/




**Shock Metamorphism, Fig. 1** Transmitted light photomicrographs of shock features in lunar anorthite, adapted from Xie et al. (2021). PPL = Plane polarized light; XPL = Cross-polarized light. (**a**) undulatory extinction observed in lunar sample 69,955,27; (**b**) Deformed twins observed in lunar sample 60,618,4; (**c**) Mechanic twins observed in lunar sample 14,303,46 which can be mistaken as planar features; (**d**) Diaplectic plagioclase glass observed from the Mistastin Lake impact structure, showing complete isotropic nature under cross-polarized light

mottled extinction pattern in cross-polarized transmitted light, caused by moderate-to-large misorientation of the crystal lattice as a result of plastic deformation. Kink bands and planar microstructures are commonly associated with optical mosaicism, as in Fig. 1b. Because a similar extinction pattern can also be developed by endogenic processes, it cannot be used as a diagnostic indicator of shock metamorphism (Ferrière and Osinski 2013).

## Planar Microstructures

Planar microstructures have long been accepted as diagnostic indicators of shock, specifically in quartz, and can be divided into planar fractures (PFs) and planar deformation features (PDFs) (e.g., Stöffler and Langenhorst 1994; Grieve et al. 1996). PFs and PDFs are distinct from each other in that PFs are open cracks, often >3 μm wide and spaced ~15–20 μm apart. Planar deformation features (PDFs), on the other hand, are closed parallel planar lamellas that are amorphous, they are less than 2 μm thick, and typically spaced 2–10 μm apart. PDFs in quartz occur as sets of closely spaced lamellae and frequently occur in multiple orientations per grain. As quartz is not common on the surface of the Moon, it is less useful as a shock indicator for lunar samples.

It is notable that while PDFs have been proposed to form in plagioclase feldspar and reported

in early works with Apollo samples as "shock-induced lamellae" of low refractive index and low or no birefringence (e.g., Short 1970a; French and Koeberl 2010), recent studies of shocked plagioclase from terrestrial impact structures and Apollo samples have failed to document any unequivocal PDFs (e.g., Pickersgill et al. 2015a, b; Xie et al. 2020). This might be due to various reasons, including misidentification (microlites, twin planes, and exsolution lamellae can easily be mistaken for PDFs, e.g., mechanical twins in Fig. 1c), structural controls relating to the crystal structure of feldspar with different compositions, and/or the presence of existing planes of weakness in the form of twin and cleavage planes (e.g., Pickersgill et al. 2015a, b; Xie et al. 2020). Further studies are warranted on the pressure-temperature conditions for forming PDFs in feldspar.

### Diaplectic Glass

The word "diaplectic" is derived from the Greek word *diaplesso*, which means to destroy by striking. Diaplectic glass refers to the optically isotropic grains that retain the chemical composition, morphology, texture, and internal features of the original mineral crystals. Unlike normal glass formed from melting, there are no flow textures or vesicles in diaplectic glass (French and Koeberl 2010). Diaplectic glass is the most characteristic shock effect in shocked plagioclase, also named as maskelynite when found in meteorites, known to preserve grain boundaries, cleavage, and twin lamellae of their precursor grain. Diaplectic glasses typically demonstrate decreasing refractive index and density with increasing shock level. They have higher density and refractive index than compositionally equivalent fused glass, which suggests that diaplectic glasses have more compact structures (Stöffler 1972; Ostertag 1983; Fritz et al. 2005). While diaplectic glass has been commonly found in terrestrial impact structures and meteorites (e.g., diaplectic glass from Mistastin Lake impact structure in Fig. 1d), it has long been noted that diaplectic glass is rare in Apollo samples with only ~1% diaplectic glass-bearing lunar basalts and anorthosites (Chao et al. 1971; Rubin 2015; Pernet-Fisher et al. 2017). In addition, it is not uncommon to find only part of grain which has become diaplectic glass in Apollo samples, leaving the remainder birefringent (Xie et al. 2021). This observation has been explained as potentially being due to sampling biases and a more global sampling is needed.

### High-Pressure Polymorphs

High-pressure polymorphs refer to high-pressure phases formed under certain pressure-temperature conditions as they do under high static pressures produced in laboratory experiments or deep in Earth's crust, such as coesite and stishovite derived from quartz in terrestrial impact structures, diamond derived from graphite, and ringwoodite and majorite from olivine and enstatite, respectively, occur in meteorites (Tomioka et al. 2017). However, not many high-pressure polymorphs have been reported in returned lunar samples except for one high-pressure polymorph of silica called stishovite, which might be due to the small sampling range (Kaneko et al. 2015).

## Methods to Study Shock Features

In addition to standard optical microscopy, advanced modern analytical techniques have also been used to study the shock features in various minerals, especially quartz and feldspar, including X-ray Diffraction (XRD), cathodoluminescence (CL), micro Fourier transform infrared (FTIR), and Raman spectroscopy (e.g., Pickersgill et al. 2015a; Martin et al. 2017; Pernet-Fisher et al. 2017; Xie et al. 2020).

### X-Ray Diffraction

X-ray diffraction method is efficient to quantify the level of crystal deformation through the measurement of streak length on 2D General Area Detector Diffraction System (GADDS) images. From undeformed crystal to amorphous, lengthening along the Debye rings (chi $\chi$ direction) progresses from single equant spots (undeformed), to short streaks, to longer streaks, to short rows of spots (asterism) with increasing pressure, ultimately to full rings (polycrystalline due to pulverization) or amorphous bands (due to

pressure- related amorphization), suggesting the increased strain causes peak broadening in the 2-theta (2θ) direction. This is due to the progressive deformation of the crystal lattice as shock level increased and the resultant variation in d-spacing of the crystal. This method has been successfully used to quantify the level of strain-related mosaic through measurement of the full width at half maximum (FWHM) of streaks in degrees chi and correlation with optically shock features in shocked plagioclase, enstatite and olivine (Hörz and Quaide 1973; Izawa et al. 2011; Pickersgill et al. 2015a).

### Cathodoluminescence (CL)

CL is the emission of photons of ultraviolet (UV) to infrared (IR) wavelengths from a material stimulated by an incident electron beam, occurring by an excited level to ground state transition of electrons in specific lattice defect. CL microscopy and spectroscopy are powerful techniques for detection of lattice defects and trace elements and analysis of their distribution, making it very unique to investigate shock-induced defects and deformation in minerals (Hamers and Drury 2011; Kayama et al. 2012; Pittarello et al. 2015). Cathodoluminescence (CL) of shocked plagioclase feldspar show a decreasing luminescence with increasing amorphization and the normal band emission at ~550–570 nm shifts to ~630 nm. Also, diaplectic plagioclase glass produces a broad band around 350 nm. In contrary to plagioclase, alkali feldspar shows increased CL intensity with increasing shock pressure (e.g., Kayama et al. 2012, 2018; Pittarello et al. 2020). However, more work on the correlation between shock intensity and the intensity of the luminescence is needed for quantifying shock levels.

### Infrared Spectroscopy

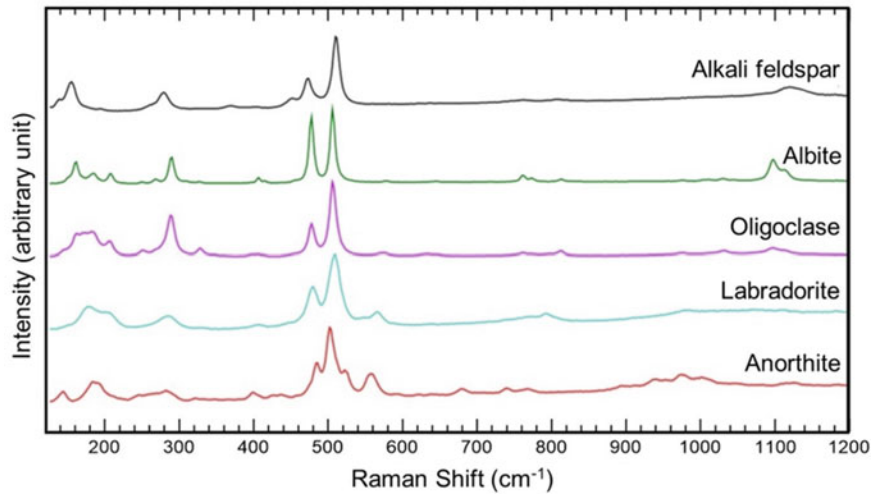
The mid-infrared region (~500–3000 $cm^{-1}$, equivalent to 3.5–20 μm) of the electromagnetic spectrum is sensitive to both the chemistry and structure of minerals. More specifically, the vibration bands that are diagnostic of mineral composition are present in the Reststrahlen band (RB) region of the spectrum, generally between 6 and 25 μm. As such, infrared spectroscopy techniques (e.g., FTIR) are useful for determining changes to crystal lattices as a result of shock metamorphism. Mid-infrared spectral characterization of the plagioclase show the Si-O diagnostic bands are the result of stretching of the Si-O bonds within the crystal structure and are situated between ~900 and 1200 $cm^{-1}$ (~11.1 and 8.3 μm). With increasing shock levels for experimentally shocked plagioclase feldspars, Si-O diagnostic bands broaden and have decreased intensities (e.g., Johnson 2012).

### Raman Spectroscopy

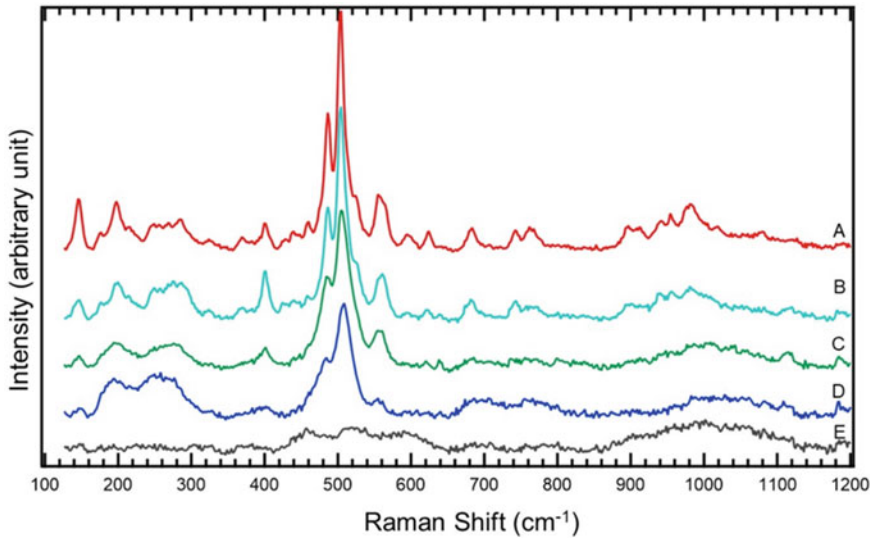
Raman spectroscopy is efficient in identifying minerals without sample preparation due to the signature Raman peaks from vibration bonds in the targets (Wang et al. 1995). The most characteristic Raman features of plagioclase are two or three Raman peaks between 460 and 515 $cm^{-1}$, as in Fig. 2. These two strongest bands correspond to motion of the oxygen atom along a line bisecting the T-O-T angle (T = Si or Al). The peaks in the 400–600 $cm^{-1}$ region are characteristic of the four-membered rings of tetrahedra in the feldspar structure, and the ones below 400 $cm^{-1}$ are from lattice vibration (Sharma et al. 1983). When examining shock features from unshocked crystals to undulatory extinction to partially isotropic regions, the FWHM of the two main diagnostic Raman peaks increase while the intensity of peaks under 400 $cm^{-1}$ decreases, suggesting the deformation of the crystal lattice. Raman spectra is also efficient in identifying diaplectic glass, the spectra displays a very broad hump with weak intensity instead of sharp narrow peaks from crystalline area, indicating the amorphous nature of diaplectic glass, as in Fig. 3. Also, Raman features are helpful in identifying microstructures as twins, planar features, and inclusions (Xie et al. 2020, 2021).

### Dynamic Shock Experiments

To help understand the natural impact process and to calibrate the mineralogical changes, dynamic shock experiments also have been conducted to reproduce shock features under controlled pressure and temperature conditions (Stöffler 1972, 1974; Hörz and Quaide 1973; Jaret et al. 2018).




**Shock Metamorphism, Fig. 2** Raman spectra of plagioclase and alkali felspar minerals, adapted from Xie et al. 2020




**Shock Metamorphism, Fig. 3** Raman spectra of shock features in lunar anorthite, showing progressive changes as shock level increase, adapted from Xie et al. 2021. (**a**) Raman spectra collected from low fractured area in 15,415, 90; (**b**) Raman spectra collected from fractured area in 60,025,230; (**c**) Raman spectra collected from slightly undulatory extinction area in 60,015,114; (**d**) Raman spectra collected from heavily undulatory extinction area in 69,955,27; (**e**) Raman spectra collected from fully extinction area (diaplectic glass) in 79,155,58

Plagioclase-rich rocks with various compositions have been experimentally shocked up to 120 GPa and the recovered samples were also investigated using advanced analytical techniques to document structural changes as a function of composition and pressure. These results help reveal the pressure plagioclase transforms to the amorphous diaplectic glass. Shock recovered samples are often heterogeneous. It is notable that no PDFs have been observed, which is likely due to the variability of the shock wave topology combined with differences in crystal orientation and composition (e.g., Johnson 2012; Jaret et al. 2018; Pittarello et al. 2020). Raman results of

experimentally shocked samples also show systematic changes similar to natural samples, including band broadening, reduction of intensities, and peak losses as shock pressure increases (e.g., Jaret et al. 2018; Fritz et al. 2019), making Raman spectroscopy an essential tool for identifying shock levels and choosing samples in future sample return missions.

## Progressive Shock Metamorphism

The progressive crystallinity changes and structural deformation in both natural and experimental shocked samples have made it possible to recognize and calibrate different levels or stages of shock metamorphism. As such, the principle of progressive shock metamorphism has been proposed and was first defined for quartzofeldspathic (felsic) rocks, including granites and gneisses in 1960s (Stöffler, 1966; Chao 1967). Since then different systems have been proposed for various terrestrial rocks, lunar rocks, and meteorites. Currently, the most up-to-date individual classification systems are focused only on rock types independently of their source planets or planetoids. As the degree of shock metamorphism, at a given shock pressure, depends primarily on the mineralogical composition and the porosity of a rock or sediment sample, different shock classification systems are required for different types of planetary rocks and sediments (Stöffler et al. 2018).

In the case of lunar rocks, the discovery of large specimens of anorthosite in the Apollo samples suggested that coarse-grained rocks consisting almost entirely of plagioclase may comprise a large part of the lunar highlands (Wood et al. 1970; Smith et al. 1970). In the last two decades, globally distributed exposures of anorthosite on the Moon have been confirmed and reported based on remote sensing data (e.g., Ohtake et al. 2009; Donaldson et al. 2014). Therefore, only the progressive stages of shock metamorphism for plagioclase-rich mafic rocks is listed here as in Table 1; for detailed references for experimental data and shock classification system of other rock types see Stöffler et al. (2018). It should be noted that natural conditions of shock metamorphism cannot be directly simulated in shock recovery experiments with rather short pressure pulse lengths, thus more work is needed for outlining pressure-temperature range of different shock levels more precisely.

## Future Directions

The Moon is the premier location for studying the crater formation process due to the lack of typical degradation processes, apart from the impact events themselves, providing valuable

**Shock Metamorphism, Table 1** Definition of progressive stages of shock metamorphism for plagioclase-rich mafic rocks

| Shock stage | Shock pressure (GPa) | Post-shock temperature (°C) | Diagnostic shock effects in plagioclase and pyroxene |
|---|---|---|---|
| S1 Unshocked | <5 | <50 | Sharp optical extinction of minerals; irregular fractures |
| S2 Weakly shocked | 5–26 | 50–220 | Plagioclase with undulatory extinction; pyroxene with mechanical twinning |
| S3 Moderately shocked | 26–28 | 220–300 | Plagioclase partially isotropic and with pdf; weak mosaicism in plagioclase and pyroxene |
| S4 Strongly shocked | 28–45 | 300–900 | Diaplectic plagioclase glass (maskelynite); strong mosaicism and pdf in pyroxene |
| S5 Very strongly shocked | 45–60 | <1850 | Plagioclase glass with flow structures and vesiculation; strong mosaicism and pdf in pyroxene; mixed melt of plagioclase and pyroxene |

information on the evolution of early earth-moon system. To better understand how impact events modify, redistribute, and mix planetary crustal materials, more lunar samples need to be collected more systematically from various impact craters on the Moon, covering the whole suite of shock features, especially moderately to strongly shocked feldspar samples which may bear more PDFs and diaplectic glass. Also, modern analytical methods need to be used more widely to study shock metamorphism in feldspar and other minerals such as pyroxene in terrestrial and lunar samples to develop efficient in-situ methods to identify different shock levels with minimal sample preparation, to enhance field experience and future exploration performance as landers and astronauts return to the Moon.

## Cross-References

- ▶ Lunar Landscape, Impact Craters
- ▶ Lunar Meteorites
- ▶ Lunar Regolith: Materials
- ▶ Impact Processes on the Moon

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